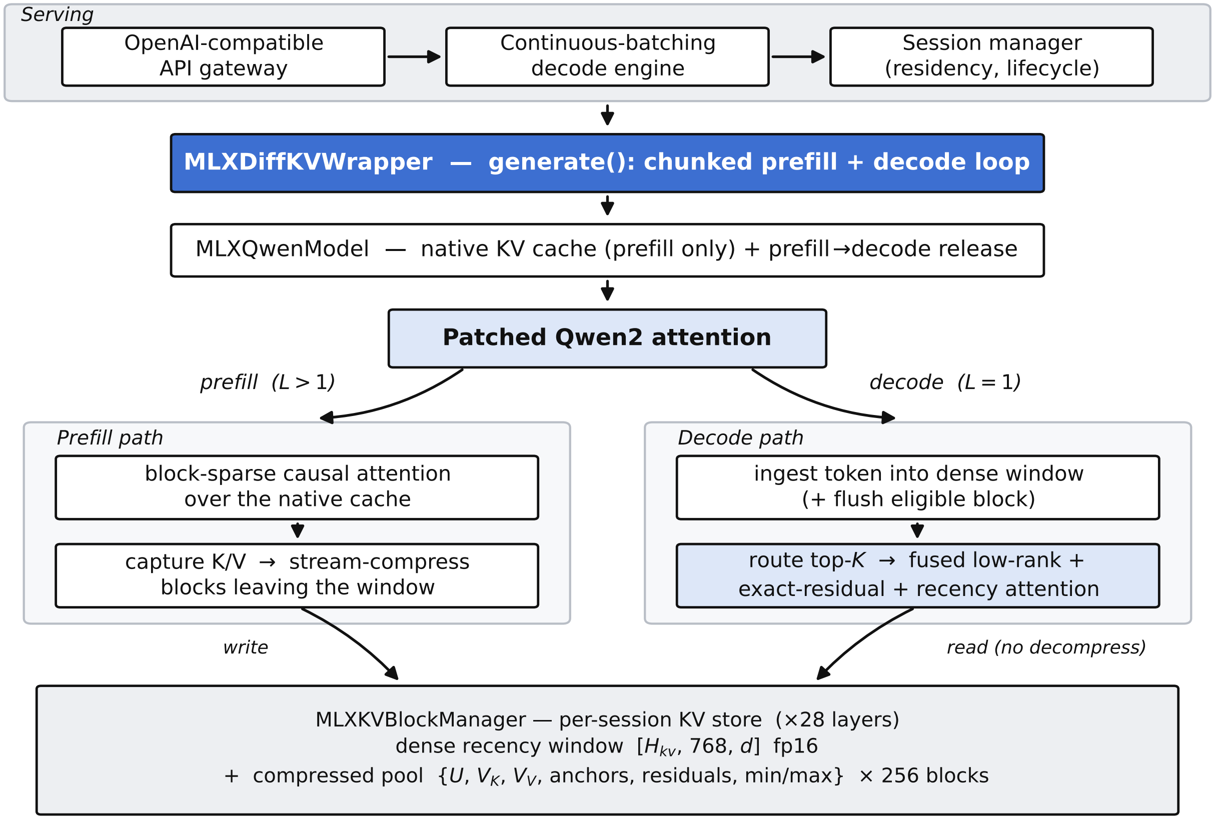

Serving
OpenAI-compatible API gateway
Continuous-batching decode engine
Session manager (residency, lifecycle)
MLXDiffKVWrapper — generate(): chunked prefill + decode loop
MLXQwenModel — native KV cache (prefill only) + prefill→decode release
Patched Qwen2 attention
prefill ($L > 1$)
decode ($L = 1$)
Prefill path
block-sparse causal attention over the native cache
capture K/V → stream-compress blocks leaving the window
Decode path
ingest token into dense window (+ flush eligible block)
route top-K → fused low-rank + exact-residual + recency attention
write
read (no decompress)
MLXKVBlockManager — per-session KV store (×28 layers)
dense recency window $[H_{kv}, 768, d]$ fp16
+ compressed pool $\{U, V_K, V_V$, anchors, residuals, min/max$\}$ × 256 blocks